// **Exercise 7: Financial Forecasting**

Program:

```java
import java.util.Scanner;
public class FinancialForecaster {
   public static double forecast(double amount, double rate, int years) {
      if (years == 0) {
         return amount;
      }
      return forecast(amount * (1 + rate), rate, years - 1);
   }
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter Initial Amount: ");
      double amount = sc.nextDouble();
      System.out.print("Enter Annual Growth Rate: ");
      double rate = sc.nextDouble() / 100;
      System.out.print("Enter Number of Years: ");
      int years = sc.nextInt();
      double futureValue = forecast(amount, rate, years);
      System.out.printf("Forecasted Value after %d years: %.2f%n",
            years, futureValue);
      sc.close();
   }
}
```

Output: